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Evaluation of a mechanistic model that estimates feed intake, growth and body composition, nutrient requirements, and optimum economic response of broilers

M.P. Reis [a,1], R.M. Gous [b], L. Hauschild [a], N.K. Sakomura [a,*]

[a] Faculdade De Ciências Agrárias e Veterinárias, Unesp Univ Estadual Paulista, Jaboticabal, São Paulo 14884-900, Brazil
[b] School of Agricultural, Earth and Environmental Sciences, University of KwaZulu-Natal, Carbis Road, 3201 Scottsville, Pietermaritzburg, South Africa

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ABSTRACT

Efficient meat production is crucial in addressing global market demands and sustainability goals. Modeling production systems has gained worldwide attention, offering valuable insights for predicting outcomes and optimizing economic returns. In the poultry industry, researchers have developed mathematical models to predict animal performance and maximize profits. These models incorporate theories to explain real-world processes and enable future event predictions. One such model is the Broiler Growth Model (**BGM**), which serves as a predictive tool for estimating feed intake, growth, and body composition of broilers. The BGM takes into account the genetic potential of the broilers, the feed they are provided, and several constraining factors that may prevent the animal from achieving their genetic potential. To evaluate the BGM, a series of simulations were performed: (i) model behavior was evaluated by simulating the response of males and females from 22 to 35 d to feeds differing in dietary protein content and nutrient density; (ii) model prediction was evaluated using the results of a protein response trial conducted at UNESP in which six dietary protein levels were fed to male and female broilers over a 56 d period; and (iii) model optimization was used to maximize economic returns in the above trial. The model behaved as expected when feeds differing in protein content were fed, with feed intake per kg of BW increasing as protein level was decreased, resulting in lower gains and higher body lipid contents. Increasing nutrient density resulted in higher feed intake in the second level, followed by a reduction in feed intake in the highest nutrient feed. The simulated response to nutrient density resulted in increasing body lipid deposition as the nutrient density increased. In comparing the simulated and actual results of the protein response trial, the overall error of prediction was up to 15% for feed intake, BW, and body protein. The optimization routine allows the simulation of different economic scenarios, helping in decision-making. The Broiler Growth Model emerges as a valuable tool for the poultry industry, offering predictive capabilities and economic optimization potential. While minor discrepancies between simulated and actual results exist, the BGM holds significant promise for enhancing efficiency and profitability in broiler production, contributing to the broader goals of sustainable broiler meat production.
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Implications

Predicting the response of broilers to different feeds and environments may guide the decisions, improving the profitability and sustainability of the production system. The Broiler Growth Model estimates the growth and nutrient requirements according to feeding program and environmental scenarios. An assumption that birds eat to achieve their genetic potential is made, and factors that prevent the animal to consume enough feed may reduce the growth. An optimization routine was included, accounting for the changes in the broiler's intake and predicting the feed program that maximizes the profit in a broiler production system.

Introduction

Nowadays, improving the efficiency of meat production has been an obligation imposed by the global market to meet the sustainability aspects (cost, environmental and social). In this scenario, modeling the production system has gained attention worldwide, and the opportunity in predicting events and optimize

* Corresponding author.
E-mail address: nilva.sakomura@unesp.br (N.K. Sakomura).
[1] Present address: Poultry R&D, Trouw Nutrition, 45215 Toledo, Spain.

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economic returns becomes attractive. For decades, researchers have developed mathematical models for the poultry industry (Fisher, 2013), attempting to reasonably predict the animal performance of a given nutrient under a given condition. Besides predicting performance, more important has been to develop frameworks that allow maximizing the profit in an agricultural system (Morris, 2006). To achieve this goal, most mathematical models rely on theories that have the purpose to explain how the real world works; thus, the theories could be compiled into empirical or mechanistic models to predict future events.

Optimizing feed for economic return is an important step toward profit maximization. To do that, Gous (2015) proposed an optimization routine that could be used as a decision tool to increase profit. In the proposed routine, a minimum-cost feed formulator was implemented in a growth model that, among other outputs, also predicts feed intake. Depending on the economic scenario, the requirement of nutrients necessary to maximize the performance may not result in maximum economic return, highlighting the importance of integrating a feed formulation tool with a growth model, allowing for predicted levels of nutrients that are economically viable, and improving the productivity of the farm.

A growth model (**BGM**) that predicts feed intake, growth, and nutrient and energy requirements for broilers was developed at Sao Paulo State University (UNESP), Agricultural Sciences and Veterinary College, Jaboticabal-Brazil. A feed formulation tool was integrated into BGM to implement an optimization routine. The objective of this study was to introduce a new version of the BGM that includes a minimum-cost feed formulator and an optimization routine that allow users to design a feed program for maximum economic return. The model was evaluated by comparing the observed responses of broilers to dietary balanced protein with those simulated in the model.

Material and methods

Theoretical assumption of the model

The proposed model applies the theory of feed intake proposed by Emmans (1981), where the author stated that "*an animal has a potential rate of normal growth at a given time and it seeks to eat an amount of a given feed which will allow this to be achieved. Its degree of success depends on the feed offered and the environment within which it is kept.*" In this concept, when energy is the first limiting resource, it is assumed that the animal will try to eat the amount of feed to supply its energy requirement. The same logic is applied for protein and essential amino acids. This theory raises the importance of describing growth, especially the growth of the main chemical components of the body. The requirement of nutrients necessary to fulfill such growth can be predicted.

The driving force to predict the genetic potential is defined by body protein deposition, and then, lipid, water, and ash are predicted from body protein using allometry equations. It is assumed that the animal consumes to meet the requirement of the first limiting component of the feed, resulting in desired feed intake (Emmans, 1987). Therefore, it seems reasonable to propose that the feed intake and nutritional requirements of an animal can be estimated as a function of its growth and body composition. Because of the difference in protein composition between the feather-free body and feather (Stilborn et al., 2010), and the differences in the growth ratio (Reis et al., 2018), the growth of each compartment is calculated separately.

In addition, the BGM takes into account factors that may prevent the animal from achieving its desired feed intake, reducing the growth. Among those factors, feed quality and environment are accounted for. A set of assays conducted at UNESP-Jaboticabal, Brazil, investigated the bulk capacity of broiler chickens and how it could constrain the feed intake based on the water-holding capacity of the feed (Nascimento et al., 2020). Environmental effect is also accounted for by BGM, assuming that birds have a maximum and minimum loss of heat, taking into account the interaction of animal heat loss with the environment (Emmans, 1989; Ferguson, 2006). In addition, the maximum lipid deposition can also constrain the energy intake and, consequently, feed intake. Due to the limited information available in the literature, the maximum lipid deposition utilized was 0.35 times of total BW, as suggested by Emmans (1981).

Model updates

The BGM is based on the AVINESP model, previously described by Hauschild et al. (2015), and further details about this model can be found in this publication. Thus, only the driven force equations, updates, and parameter values will be described here. Several studies have been conducted at UNESP-Jaboticabal, Brazil, to update the model parameters regarding the genetic potential of broilers. Allometry equations were also included to predict parts of broiler chickens based on studies conducted by Azevedo et al. (2021a). As the bulk capacity of broilers' intestines constrain feed intake, one study was conducted to evaluate such an effect (Nascimento et al., 2020). Finally, a feed formulator tool and an economic optimizer routine were implemented in the software, allowing to formulate feeds that maximize the economic return.

Genetic growth potential

The BGM is a dynamic model and starts calculating the daily genetic potential for growth, considering protein, lipid, water, and ash depositions on the feather-free body and protein deposition on feathers. The first derivative of the Gompertz function (Gompertz, 1825) is used to estimate the daily protein deposition (P_d), as demonstrated in Eq. (1).

$$P_d = \text{maturity rate} \times \text{protein weight} \times \ln(\text{maturity protein/protein weight}), \quad \text{g/d} \tag{1}$$

where maturity rate is per day, protein weight and maturity protein are in g. This equation demonstrates that the growth rate is dependent on the current status of the animal. The growth parameters were updated based on the study published by Gonçalves et al. (2020, Table 1).

Lipid deposition is dependent on nutrition, and the prediction of body lipid deposition is performed as suggested by Emmans (1981). The author proposed that animals have a desired lipid deposition, by relating a so-called "normal" lipid content with body protein. It was assumed that a "normal" body lipid content (dL_c) has a relation to protein and the lipid:protein ratio at maturity (LP_m) was used to describe this relation (Emmans, 1997).

$$dL_c = LP_m \times \text{maturity protein} \times (\text{protein weight/maturity protein})^b, \quad \text{g/d} \tag{2}$$

The value of b is estimated following the equation proposed by Emmans (1997):

$$b = 1.46 \times LP_m{}^{0.23}, \quad \text{g/d} \tag{3}$$

Thus, desired body lipid deposition (dL_d) is determined in function of protein (Emmans, 1987):

$$dL_d = \text{protein deposition} \times LP_m \times (b) \times (\text{protein weight/maturity protein})^{b-1}, \quad \text{g/d} \tag{4}$$

The set of Eqs. (1)–(4) is particularly interesting since *Pt* is the only state variable. To predict lipid deposition in situations where the energy is over- or under-consumed, an equation is added to the model to compensate for the desired body lipid deposition.

$$\text{Lipid deposition} = dL_d + (dL_c - \text{lipid weight})/1000, \quad (\text{g/d}) \qquad (5)$$

Water and ash contents in the body were demonstrated to be correlated with protein (Emmans, 1989, Ferguson, 2006, Gous and Berhe, 2006), hence, for simplistic reasons, allometric equations ($\ln(Y) = a + b \times \ln(Pt)$) were proposed to estimate the deposition of body water and body ash, where Y is the deposition of water or ash in the body. A spreadsheet was elaborated to demonstrate how the equations were combined (Supplementary Material S1). It can be used as a template to estimate the growth potential and the desired feed intake for a growing broiler.

The prediction of the commercial body parts was implemented into BGM to estimate breast, drum, drumstick, and wing weights. Allometric equations were used to estimate the weight of each component according to body protein weight (Reis et al., 2022). All the parameter values used in the model are given in Table 1.

Maximum feed intake capacity

A study was conducted at UNESP, Jaboticabal-Brazil (Nascimento et al., 2020), to determine the maximum feed intake of broilers for a particular feed. Physical and chemical aspects of the feed were used to describe the maximum feed intake per kg of metabolic BW ($kg^{0.67}$). The water-holding capacity of the feed (g of water per g of feed) was demonstrated to have the most significant impact on maximum intake. A quadratic equation was given to describe the maximum intake of food per kg of metabolic BW ($g/kg^{0.67}$), namely maximum scale feed intake (**SFI**$_{max}$):

$$\text{SFI}_{max} = 369.4 - 115.7 \times \text{water holding capacity} + 11.96 \times (\text{water holding capacity})^2, \quad (\text{g/kg}^{0.67}) \qquad (6)$$

Feed formulator and economic optimizer

A feed formulator was developed using the Simplex algorithm (Dantzig and Thapa, 2003) and integrated into BGM. The objective set in the linear program optimization algorithm is to minimize the cost of the feed, given a set of constraining inputs (minimum and maximum inclusion of ingredients and nutrients in the feed formula). As Azevedo et al. (2021b) demonstrated, broilers' responses to nutrient intake might change between dietary levels; therefore, minimizing the feed cost may not represent the maximum economic return. For that, it is necessary to account for the shift in the broiler responses to a feed change. As mentioned, the BGM simulates the response of broilers, considering the feed given. Thus, the economic optimizer is a tool developed to integrate the growth model and the feed formulator (Gous, 2015), optimizing a feeding program based on the feed's energy levels or amino acids. A routine was developed where the user informs a range of the feed component (metabolizable energy, balanced protein, standardized ileal digestible lysine), and the feed formulator produces all possible feeding programs within the range informed, which is simulated on the growth model. This routine is automatically performed, and the feed program that maximizes the profit is displayed at the end of a simulation sequence.

Model behavior

To illustrate the behavior of the broiler growth model, we simulated five different feeds for Ross 308 chicks aged 22–35 days by adjusting dietary energy and essential amino acids. Three feeds were formulated with a consistent metabolizable energy level (3 100 kcal/kg) and varied levels of SID-Lys (0.95, 1.05, and 1.15%). The remaining two feeds were designed to contain either 3 400 or 2 800 kcal/kg of metabolizable energy and 1.15 or 0.95% SID-Lys. It was assumed that broilers were growing at their potential before day 22 (Table 2). Essential amino acids were maintained at a constant level with dLys, and all other nutrients were adjusted to meet or exceed the nutritional requirements of broilers, as per Rostagno et al. (2017). Temperature, relative humidity, and air velocity were set according to the guidelines recommendation (Aviagen, 2022).

Table 1
Parameters used in the Broiler Growth Model.

Parameters[1]	Feather-free Body	Feather
P_m	1.266	0.217
B	0.039	0.039
LP_m	1.4	–
a_{Water}	1.987	–
b_{Water}	0.039	–
a_{ash}	−1.419	–
b_{ash}	0.942	–

Abbreviations: Pm = protein at maturity (kg); B = maturity rate (/day); LPm = lipid to protein ratio at maturity (kg/kg); a and b = the constant and regression terms of the allometry equation for water and ash in the body.
[1] All parameters were obtained from Gonçalves et al. (2020).

Model evaluation

Results from a study conducted at UNESP-Jaboticabal to investigate the responses of Ross 308 broilers to dietary balanced protein were used to evaluate the BGM (Azevedo et al., 2021a). In this study, the broilers were fed with a feed program divided into four 2-week phases (1–14, 15–28, 29–42, and 43–56 days), with dietary balanced protein levels being reduced in each subsequent period, but with the same relative difference between levels being maintained. Dietary balanced protein was defined as a constant ratio of essential amino acids with lysine. Brazilian tables (Rostagno et al., 2017) were used as a reference, and the relative differences applied were 0.60, 0.70, 0.85, 1.00, 1.15, and 1.30 times the reference for standardized ileal digestible lysine. The SID-Lys ranged from 8.27 to 17.9, 6.65 to 15.8, 5.73 to 13.8, and 5.28 to 12.8 g/kg of feed, respectively, for each 2-week phase. More details about the methods and feeds used in the study are provided by Azevedo et al. (2021a).

The information about genetic strain, diet, and environment temperature was used as an input in the BGM, to simulate the response of broilers inside each of the six dietary balanced protein levels. The output values of feed intake, BW, body protein, and body lipid were compared with the observed database. The residual of the model was assessed using the mean square prediction error (**MSPE**) and its decomposition, mean bias (error central tendency – **ECT**), slope bias (error due to regression – **ER**), and random errors (error due to disturbances – **ED**), according to Pomar and Marcoux (2005). Furthermore, the square root of MSPE was calculated and expressed as a proportion of the mean predicted, named overall error. The ECT, ER, and ED were expressed as a percentage of MSPE.

Optimizing broiler production to maximize the economic return

A simulation was done to demonstrate the use of BGM software to maximize the economic return in broiler production. The feed and income costs applied in the optimization procedure were the same reported by Azevedo et al. (2021b). Then, an optimization routine was used to obtain a feed formula that maximizes the

Table 2
Dietary metabolizable energy (ME) and digestible lysine used in five feed programs simulated in the Broiler Growth Model.

Simulated diets	ME, kcal/kg	Digestible lysine, %
1	3 100	0.95
2	3 100	1.05
3	3 100	1.15
4	2 800	0.95
5	3 400	1.15

economic return, by changing the balanced protein in the feed. Saleable products were defined as live, dressed (carcass), and further processed (breast minus skin, leg (thigh and drum), and wing). Digestible lysine was used as the reference amino acid, considering the ideal ratio recommended by Rostagno et al. (2017). The simulation was performed for both males and females.

Results

Effects of energy and balanced protein on performance and body composition - simulation

In the simulation on BGM, the feed intake of broilers consuming crescent levels of balanced protein and constant energy (Fig. 1a) increased by around 1% when digestible lysine increased from 9.5 to 10.5 g/kg. On the other hand, a slight decrease in feed intake (0.03%) was observed when the digestible lysine increased from 10.5 to 11.5 g/kg of feed. The BW increased by around 3% when digestible lysine was increased from 9.5 to 11.5 g/kg. The difference in BW between feeds containing 10.5 and 11.5 g of digestible lysine/kg of feed was around 0.2%. The changes in body composition were highly affected by dietary balanced protein (Fig. 1b). Body protein increased by 4.3% when digestible lysine increased from 9.5 to 11.5 g/kg, while the increase was only around 0.3% when digestible lysine increased from 10.5 to 11.5 g/kg. Body lipid was also affected by the dietary balanced protein, where the lowest level of digestible lysine produced the highest lipid content, which was 5 and 6% higher in comparison with broilers consuming 10.5 and 11.5 g of digestible lysine/kg of feed, respectively.

The simulations performed with increasing nutrient density (Fig. 1c and d) demonstrate a different behavior in comparison with the scenario where only dietary balanced protein changed. The lowest feed intake was observed for broilers consuming the highest density feed (3 400 kcal/kg of ME and 11.5 g/kg of dLys) while the highest value is observed in the feed (3 100 kcal/kg of ME and 10.5 g/kg of dLys), representing an increase of 5%. Compared with broilers consuming the highest density feed, the feed intake increased by 3% when broilers were fed a feed with 2 800 kcal/kg of ME and 9.5 g/kg of dLys. The BW was 1 and 7% higher when broilers were fed with 3 400 kcal/kg of ME and 11.5 g/kg of dLys, than the broilers fed the second and the third lowest level of nutrient density, respectively. The body composition followed similar results to the BW, but the proportions of change among protein and lipid content were markedly different. Body protein was reduced by around 1 and 5%, respectively, as the nutrient density reduced, while body lipid was reduced by 7 and 26%.

Optimizing a feed program based on balanced protein

The dietary balanced protein level necessary to maximize the economic return depends on the sealable product and, to a lower extent, by gender and age (Table 3). When male broilers are commercialized at 42 days old and the live bird is the final product, the dLys to maximize profit is 8.5 g/kg, increasing the dLys by around 14 and 21% for dressed and further processed birds, respectively. For broilers sold at 56 days of age as a live bird, the dLys to maximize profit is 8.3 g/kg. An increase of 18 and 24% in dLys was estimated to maximize the profit of male broilers sold dressed or further processed.

Model evaluation

To evaluate the model, the results simulated in the BGM were compared to those observed in the trial conducted at UNESP-Jaboticabal and published by Azevedo et al. (2021a). The response of broilers to crescent levels of dietary BP was similar to the observed data (Fig. 2). The root MSPE values were 0.574, 0.281, 0.028, and 0.112 kg/bird for cumulated feed intake, BW, body protein, and body lipid, respectively. These results show the accuracy of the outputs generated by the BGM and are in the same unit as each response variable. The overall error of prediction was below 15.5% for cumulated feed intake, BW, and body protein, with the most errors estimated due to disturbances (79, 66, and 94%, respectively). The overall error estimated for body lipids was 42%, with the most part estimated due to disturbances (85%).

Discussion

A mechanistic broiler growth model aims to represent the reality of broiler production. In this sense, the model should be developed based on a consistent theory. The BGM was constructed under the theory that growing birds seek to reach their protein deposition potential, and predicting the nutrients necessary to reach this objective is primordial. It is also important to consider the efficiency of nutrients utilized by chickens, which allow the birds to reach their potential. To have simulated results close to reality, some factors that may prevent the animal to grow at their potential should be considered. Finally, the BGM accounts for the influences of environment, feed quality, and intestinal bulk capacity, attempting to predict the actual feed intake and, consequently, the amount of nutrients available for growing.

In illustrating the behavior of the model, the simulations aimed to demonstrate the theories compiled within the model that drive the output. Fig. 1a and b provides insights into how broilers respond to a feed change, specifically focusing on the variation of balanced protein. In this scenario, as the dietary digestible lysine decreases from 1.15 to 0.95%, it becomes the first limiting nutrient in the feed, and the desired feed intake would need to increase to accommodate such changes. However, it is noteworthy that the actual feed intake decreased by approximately 1%, consequently impacting the broilers' BW by 3%. This observation indicates that broilers were prevented from consuming a sufficient amount of feed to meet their nutritional needs. Among the mechanisms that may play a role in constraining feed intake in broilers, the bulk capacity of the intestinal tract (Nascimento et al., 2020; Taylor and Kyriazakis, 2021) and maximum lipid deposition (Emmans and Fisher, 1986) can have an effect on feed intake. Therefore, even though the desired feed intake increased, the actual feed intake reduced and the broilers were unable to consume the necessary amount, leading to a reduction in their BW, particularly in their body protein weight. It is worth noting that body lipid content increased by approximately 6% between broilers consuming the lowest and highest balanced protein feeds. This observation suggests that broilers consumed more feed in an attempt to meet their first limiting component in the feed. However, this action resulted in overconsumption of energy, leading to an increase in lipid deposition. Similar findings were reported by Azevedo et al. (2021a) and Reis et al. (2022). In summary, the simulations conducted in this study provide valuable insights into the effects of feed changes

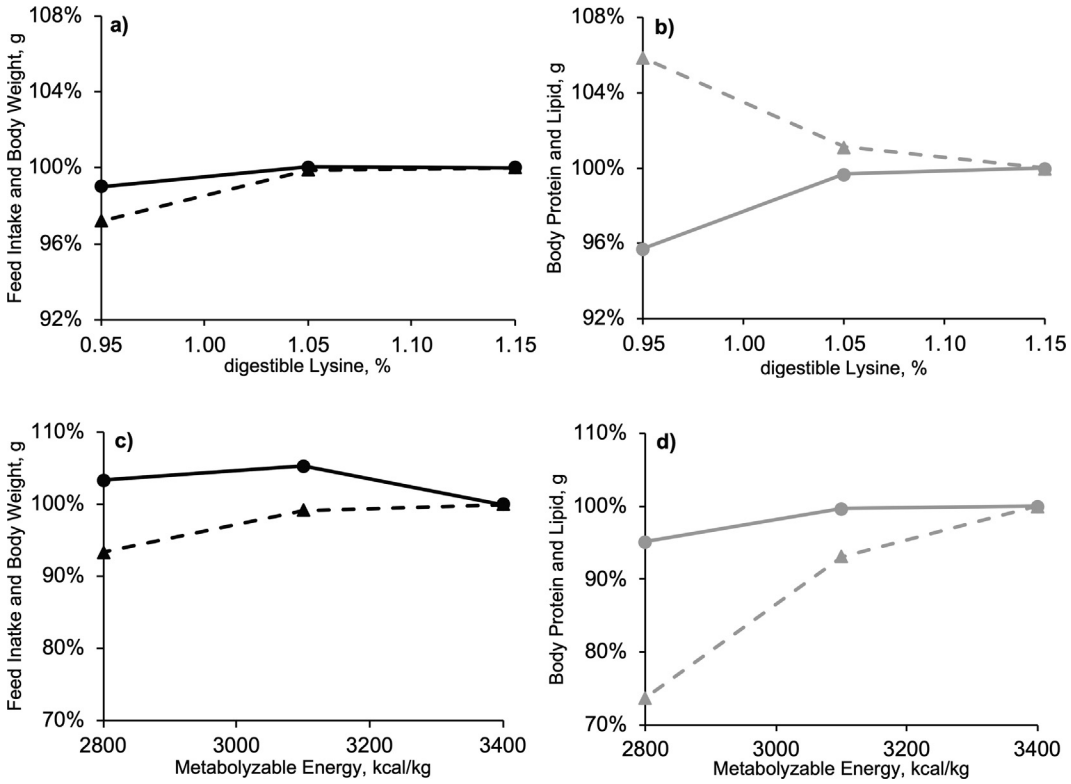


Fig. 1. Simulated feed intake (a and c, solid line), BW (a and c, dotted line), body protein (b and d, solid line), and body lipid (b and d, dotted line) of broilers consuming a feed with constant energy (3 100 kcal/kg) and increasing levels of digestible lysine in a and b, and increasing nutrient density in c and d, relative to the highest levels applied.

Table 3
Digestible lysine levels estimated in the broiler growth model (BGM) and the observed results that maximized BW and profit of whole body and selling parts of broilers.

Sales Category	Day 42				Day 56			
	BGM Model		Azevedo et al. (2021b)		BGM Model		Azevedo et al. (2021b)	
	Male	Female	Male	Female	Male	Female	Male	Female
Live Bird	0.85	0.85	0.92	0.88	0.83	0.82	0.88	0.82
Dressed bird	0.97	0.97	0.97	0.94	0.98	0.96	0.95	0.91
Further processed	1.03	1.01	1.04	1.01	1.03	1.01	1.02	0.99

on broiler behavior and performance. The observations align with existing literature and highlight the complex nature of feed intake regulation and nutrient utilization in broilers. Further research is warranted to delve deeper into the underlying mechanisms and identify strategies to enhance feed efficiency and broiler productivity.

The simulations presented in Fig. 1c and d indicate that feed intake increases when protein and energy are simultaneously reduced in the feed. This observation suggests that in previous simulations, maximum lipid deposition constrained the desired feed intake of broilers. However, in the simulations depicted in Fig. 1c and d, it appears that intestinal capacity plays the most significant role in constraining the desired feed intake. It is important to note that even though the actual feed intake shown in Fig. 1c did not decrease, the desired feed intake was not achieved, resulting in an under-consumption of energy and nutrients. This effect is evident in Fig. 1d, where body lipids are drastically reduced.

Before the use of a model, an evaluation should be done. The agreement between simulation results and external data enhances confidence in the model's predictions. Nevertheless, it is essential to exercise caution and recognize that even a well-performing model may have limitations or areas of improvement. Continuous evaluation is necessary to identify potential shortcomings or discrepancies between simulated and observed data. These evaluations help refine the model, enhance its accuracy, and ensure its suitability for specific applications.

According to the evaluation results, the BGM model can predict the feed intake and BW of broilers fed different energy and protein levels with good accuracy, based on the overall error of 15% for feed intake and an even lower error for BW (12.4%) and body protein (8.15%). Following the theory of Emmans (1981), broilers increase feed intake when dietary energy or nutrient is limiting, attempting to consume enough nutrients to grow at their potential. However, the bulk capacity of birds may constrain their feed intake. If broilers cannot consume the desired amount of nutrients, the growth is reduced. In the paper published by Azevedo et al. (2021a), however, a low dietary level of balanced protein resulted in reduced feed intake, but only when the deficiency became too severe. We observed the same result in the simulations performed in the BGM, as demonstrated in.

An effect in feed intake is expected when the nutrient diet level changes since birds try to compensate for the change in the feed formula (Fig. 1), which may cause a change in body composition and the economic return (Table 3). Because feed intake increases as broilers grow, the concentration of dietary protein required is reduced, but it is impractical to reduce the dietary concentration

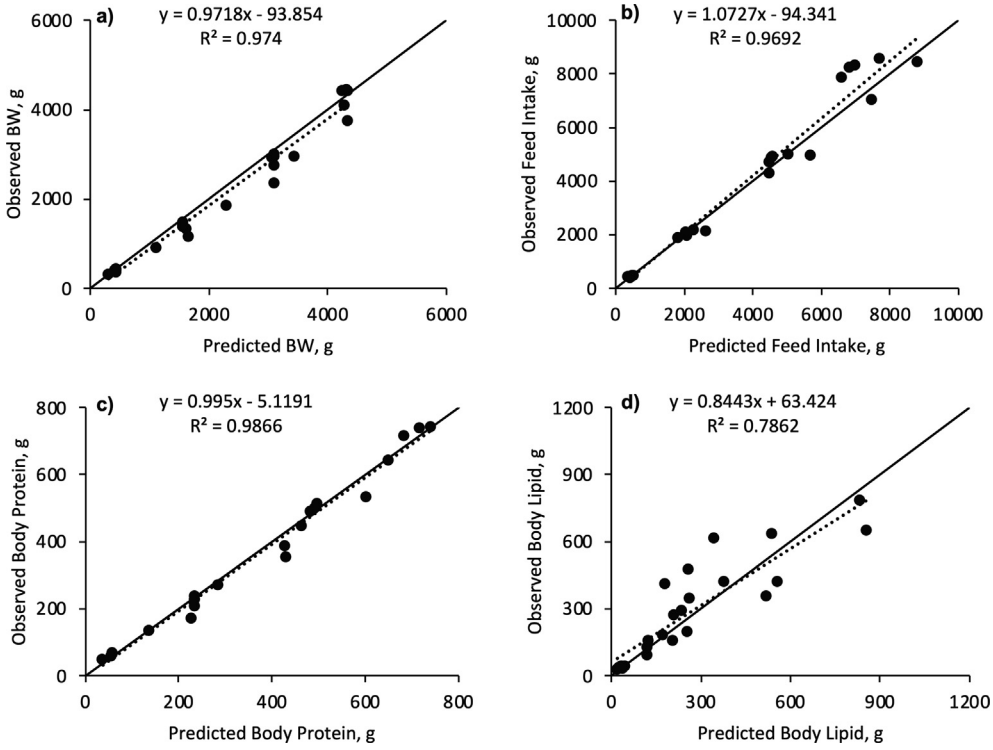


Fig. 2. Relationship between observed and predicted broilers data (N = 24) for BW (a, relative mean square prediction error [rMSPE] = 12.4%, error in central tendency [ECT] = 31.5%, error due to regression [ER] = 2.10%, and error due to disturbances [ED] = 66.4), feed intake (b, rMSPE = 15.5%, ECT = 9.26%, ER = 11.5%, and ED = 79.3%), body protein (c, rMSPE = 8.15%, ECT = 6.11%, ER = 0.180%, and ED = 93.7%), and the body lipid (d, rMSPE = 42.5%, ECT = 3.99%, ER = 10.7%, and ED = 85.3%). x = y line (solid line) and regression line (dotted line). The intercepts and slopes from the regression line did not differ from 0 and 1 ($P > 0.05$), respectively, for the response variables evaluated. Database from Azevedo et al. (2021a).

every day, hence, it is recommended a phase-feeding program. In commercial broiler production, feeding programs are usually divided into phases, using two to five diets decreasing in protein throughout the production cycle. Assuming that the birds are given ad libitum access to feed, they are fed sub-optimal levels of protein (amino acids) at the beginning and overfed toward the end of each phase. The nutrient density should consider the growth potential of the bird as well as the feeding costs and revenue; thus, the feed with a proper nutrient density is offered to broilers as demonstrated in Table 3.

At the start of each phase, birds will attempt to consume enough of the limiting resource. When the first limiting nutrient in a diet is an amino acid, broilers will overconsume energy which is deposited as body lipid (Fig. 1b). In practical circumstances, broilers are often prevented to consume a sufficient amount of feed to grow at their potential. Three mechanisms were added into the BGM attempting to account for those circumstances: 1 – when broilers cannot lose to the environment the heat produced, e.g., hot environment (Emmans, 1989); 2 – when the broilers reach their maximum intake of food due to a physical limitation in the intestine (Nascimento et al., 2020); and 3 – the energy intake should reduce when the maximum lipid deposition is reached (Reis et al., 2022). As protein is added to the feed, the undersupply of amino acids reduces and then becomes excessive compared to the requirement. At this point, the bird might be able to metabolize the excessive body lipid to obtain the energy required (Nonis and Gous, 2012), and feed efficiency is improved. In accordance with the aforementioned, the body protein is lower when broilers are fed a low protein feed and regular energy (Fig. 1b), demonstrating that simulated broilers were not able to consume a sufficient amount of feed to reach their potential for protein deposition. Furthermore, body lipid increases, since more energy is available for lipid deposition. Similar results were observed by Azevedo et al. (2021a). Growing chickens fed ad libitum with low dietary levels of amino acids increases the feed intake and the fat deposition, taking more time to reach the target slaughtering weight (Gous, 1990; Azevedo et al. 2021a).

The feed intake is dependent, among other factors, on the current state of the animal, which will influence the amount of protein and lipids that will be deposited each day (Gous, 2012). The results from the simulation of broilers consuming a feed with crescent levels of dietary energy and protein content demonstrated a small variation in body protein (Fig. 1d). In addition, the feed with 3400 kcal/kg of ME and 11.5 g/kg of dLys showed the highest lipid deposition since the birds consumed feed to achieve the requirement for lysine, overconsuming energy, which was deposited as lipid.

The economic optimizer presented in this study is a computer routine designed to optimize feed formulation in broiler production. The optimizer begins by taking the user's input of feed formulas to be optimized. These feeds are then entered into the BGM, where simulations are performed to evaluate their performance. By combining the simulation results with market prices, the optimizer calculates the revenue generated by each feed formulation. The economic optimizer incorporates an iterative process to refine the feed formulas. Using the simulation outcomes, the optimizer automatically adjusts the feed formulations, making slight modifications to the component chosen by the user. For instance, in this study, we focused on optimizing balanced protein as the selected feed component. The updated feed formulas are then fed back into the BGM for subsequent simulations. During each iteration, the economic optimizer computes the feeding cost, which is determined by multiplying the feed intake by the price of the feed. Additionally, the revenue is calculated based on the market price of BW,

carcass, and parts. This iterative loop continues until a specific stopping criterion is met. Ultimately, the feed formula that maximizes the gross margin (revenue minus feeding cost) is provided to the user as the optimal solution. In practical applications, a similar process is typically performed by a nutritionist. However, this manual approach can be time-consuming as it requires waiting for a broiler flock to be produced, analyzing the data, reformulating the feed, and repeating the cycle. The advantage of utilizing a well-developed model, as demonstrated in this study, is the automation of this process, resulting in significant time savings. By leveraging the economic optimizer, nutritionists and researchers can efficiently optimize feed formulations for broiler production, considering both revenue and feeding costs. The automated nature of the optimizer enhances decision-making processes and allows for rapid evaluation of various feed formulas.

The results illustrated in this study demonstrate how the feed formula can be managed to influence the response of broilers and optimize economic returns. In agreement with the food intake theory used in this model, as dietary balanced protein changes, the animals will attempt to consume sufficient food to meet their requirements and achieve their potential growth. By slightly reducing dietary balanced protein, it is expected that broilers will increase their feed intake (Azevedo et al., 2021a), resulting in energy overconsumption, which is stored as body lipid. Since broiler fat may have a low economic value when sold separately, this scenario might be undesirable if the broilers are further processed before being sold. However, when the producer receives payment per kg of live BW, this presents an opportunity to improve their economic return. This explained why the balanced protein content to maximize the economic return is lower when live broilers are the market option. On the other hand, if the producer receives payment per kg of carcass or cuts, the focus may lie in maximizing protein deposition and minimizing body lipid content. When higher balanced protein feed is offered to broilers, one may expect a reduction in feed intake and, consequently, in body lipid content, as similarly evidenced by Azevedo et al. (2021b). Simultaneously, the protein deposition is close to broilers' potential, maximizing meat production. As demonstrated, the decision to increase or reduce a feed component (energy or amino acid) is dynamic, influenced by economic variables and multifactorial factors such as feed composition, genetic potential, and environment. The use of predictive models can help users overcome this complex issue and improve the economic return in a production farm.

The outputs of the economic optimizer demonstrate the benefits of using a growth model together with a minimum-cost feed formulator to estimate the maximum economic return. In the simulation performed herein, the amount of dietary digestible lysine necessary to achieve maximum performance was not the same; similar results were reported by Pack et al. (2003). Other methodologies to estimate the economic level of dietary protein were developed over the years (Pesti and Miller, 1993; Eits et al., 2005); however, since the feed intake is expected to increase or reduce when the dietary protein change, an economic optimization process must consider such variations (Gous, 2015). As demonstrated herein, the BGM can predict the feed intake and BW of broiler chickens with good accuracy, accounting for the variations in the diet. The economic optimizer routine developed accounts only for the revenue and costs; however, it is necessary to be aware that other circumstances may be chosen as the final objective, such as nitrogen excretion. In this situation, a similar approach is recommended.

Conclusion

The results demonstrate that the Broiler Growth Model predicts the growth potential and actual feed intake with reasonable accuracy in comparison to observed data. The model behavior indicates that the mechanisms and theories that drive the predictions are in line with published literature. In addition, the Broiler Growth Model could be used as a decision tool in a production system, performing simulations to decide the feeding program for maximum economic return.

Supplementary material

Supplementary material to this article can be found online at https://doi.org/10.1016/j.animal.2023.101016.

Ethics approval

Not applicable.

Data and model availability statement

As an example of how to organize the equations presented herein, a spreadsheet was developed and attached as Supplementary Material S1. The model presented is a registered software (process number BR512019002861-9), and further information should be addressed to the corresponding author.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) did not use any AI and AI-assisted technologies.

Author ORCIDs

Matheus de Paula Reis: https://orcid.org/0000-0001-8255-9032.
Nilva Kazue Sakomura: https://orcid.org/0000-0001-5707-4113.
Luciano Hauschild: https://orcid.org/0000-0002-4175-3987.
Robert Gous: https://orcid.org/0000-0002-5457-3643.

Author contributions

Reis, M.P.: Conceptualization, methodology, formal analysis, writing original draft, **Sakomura, N. K.**: Conceptualization, methodology, writing reviewing, supervision. **Hauschild, L.**: Conceptualization, methodology, formal analysis, writing reviewing, **Gous, R.**: Conceptualization, methodology, writing reviewing, supervision.

Declaration of interest

None.

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Transparency declaration

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